

Lecture Notes 1

A Differential Equation is an equation that contains derivatives.

ODE: Ordinary Differential Equation ; PDE: Partial Differential Equation

By order we mean the highest-order derivative that appears in the DE.

$$\frac{dy}{dt} = y' = f(t, y). \quad \leftarrow \text{basic first-order ODE.}$$

A solution to the above ODE is a function $y = y(t)$ that reduces it to an identity given by

$$y'(t) = f(t, y(t))$$

for some appropriate domain points.

Ex// One of the simplest: $y' = 2y$, "the derivative of what function gives back twice what you started with?"

If we make a guess, $y = e^{2t}$, then plugging this into the first-order ODE gives

$$\text{LHS: } y' = 2e^{2t} \quad \text{which is an identity.}$$

$$\text{RHS: } 2y = 2(e^{2t})$$

This holds for any $t \in \mathbb{R}$.

Ex// The simplest ODE is of the form $\frac{dy}{dt} = f(t)$. The solution to this was the subject of second-semester calculus.

$$y = \gamma(t) = \int f(t) dt + C \quad \text{is the solution.}$$

Note: the "+ C" means we have more than just one solution, we have an infinite number of solutions! The entire collection of solutions is called a family of solutions.

Ex// $\frac{dy}{dt} = y^2 + 1$, $y = \tan(t + C)$ is called the general solution, when $t \in (-\frac{\pi}{2} - C, \frac{\pi}{2} + C)$

$$\frac{dy}{dt} = \sec^2(t + C) \quad \& \quad \tan^2(t + C) + 1 = \sec^2(t + C), \quad \text{where } C \text{ is any number.}$$

A particular solution arises when we pick (or are given) a particular value for C .

Sub Ex// $C = -\pi/2$

$y = \tan(t + C) = \tan(t - \pi/2)$ is a particular solution to the first-order ODE given by $\frac{dy}{dt} = y^2 + 1$.

Defi: A differential equation is called autonomous if $\frac{dy}{dt} = f(y)$, that is, the independent variable t does not appear on the RHS.

Ex// $\frac{dy}{dt} = y^2 + 1$ is an autonomous first-order ODE.

Initial-Value Problems

In order to determine the "C" value for some ODEs we need to specify a condition called an initial-value (initial since we consider t to stand for time)

Defn: The combination of a first-order ODE and an initial-value (or initial condition)

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0$$

is called an initial-value problem. Its solution will pass through (t_0, y_0) .

Note: IVPs have particular values if they exist at all.

Ex// $\frac{dy}{dt} = y^2 + 1$, $y(\frac{\pi}{4}) = 0$ gives the solution

$$y(t) = \tan(t + C)$$

$$\Rightarrow y(\frac{\pi}{4}) = 0 = \tan(\frac{\pi}{4} + C)$$

$$\Rightarrow \frac{\pi}{4} + C = 0 \quad (\text{why?})$$

$$\Rightarrow C = -\frac{\pi}{4}$$

$\therefore$ the solution to the above IVP is the particular solution: $y(t) = \tan(t - \frac{\pi}{4})$ ✓

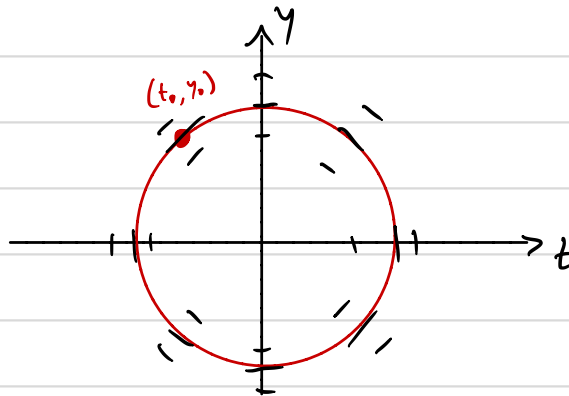
Direction Fields (Slope Fields)

Sometimes it is not possible to solve a DE explicitly, or it may be more telling to view a graphical solution to a DE.

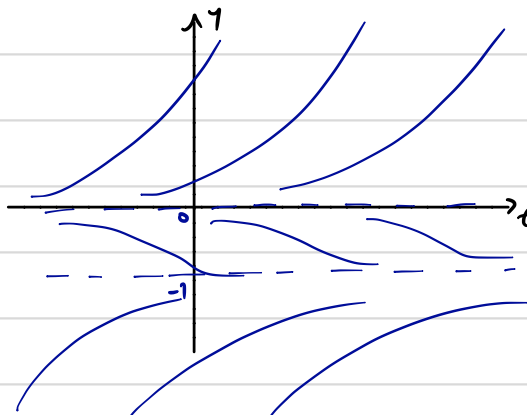
Graphically, a solution to a first-order ODE is a function whose slope at each point is specified by the derivative. *sometimes called the Tangent Space.*

The collection of slopes in the ty -plane is called a direction field (or a slope field).

Ex// #10 in §1.2 $y' = -\frac{t}{y}$ (see MATLAB presentation)
 $(t, y) \in [-2, 2] \times [-2, 2]$



Ex// #14 in §1.2 $y' = y(y+1)$ (see MATLAB presentation)
 $(t, y) \in [-2, 10] \times [-4, 4]$



What you notice is two "lines" where solution curves are diverging from or converging to!
It's also super easy to see where those lines come from.

Def'n: A solution of an ODE that does not change over time is called an equilibrium solution.

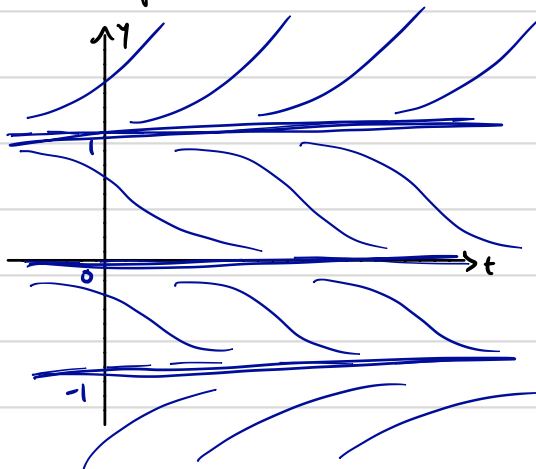
For a first-order ODE, the equilibrium solution is always of the form $y(t) = C$ which always comes from solving $y' = 0$.

Ex// $y' = y(y+1) = 0 \Rightarrow y = 0$ & $y = -1$ are equilibrium solutions!

But they look drastically different! From one equilibria the solution curves diverge, and the other they converge! Let's look at another example.

Ex// $y' = (y+1)(y)^2(y-1)$ (see MATLAB presentation)

$(t, y) \in [-2, 10] \times [-2, 2]$



From this graph we can tell three different types of equilibria that occur!

Def'n: An equilibrium solution may be stable, unstable, or semi-stable.

- It is stable if ICs near the equilibrium tend towards the equilibrium as $t \rightarrow \infty$.
- It is unstable if ICs near the equilibrium tend away from the equilibrium as $t \rightarrow \infty$.
- It is semi-stable if ICs on one side of the equilibrium tend towards it while they tend away from it on the other side as $t \rightarrow \infty$.

Ex// $y' = (y+1)(y^2)(y-1)$.

$y = -1$ is a stable equilibrium

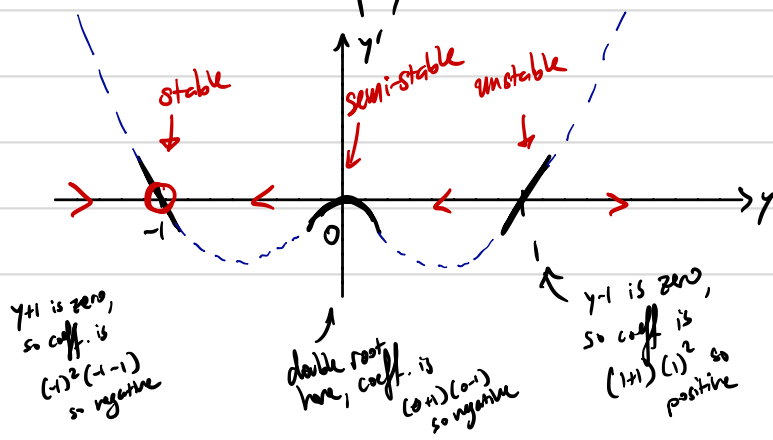
$y = 0$ is a semi-stable equilibrium

$y = 1$ is an unstable equilibrium.

It's particularly clear w/ autonomous ODEs.

Can also determine the behavior of autonomous ODEs by studying (y, y')

Ex// $y' = (y+1)(y^2)(y-1)$, zeros at $-1, 0, 1$, 0 is a double root, whole thing is pos. fourth order polynomial.



The Method of Isoclines

There is another way to draw by hand the direction fields is called the method of isoclines.

Def'n: An isocline of a first-order ODE $y' = f(t, y)$ is a curve in the ty -plane along which the slope at any point is constant.

Thus, it is given by $y' = \text{const} = f(t, y)$, or $f(t, y) = C$.

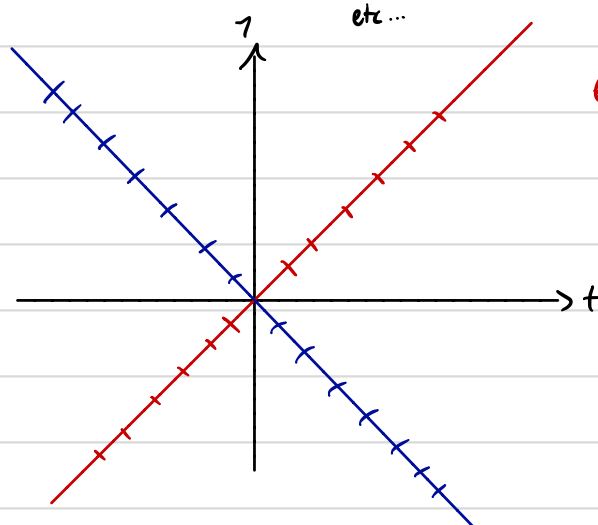
Ex// § 1.2 #10 $y' = -\frac{t}{y}$

To find isoclines, set $f(t, y) = \text{const}$.

$$\therefore -\frac{t}{y} = C, \text{ or } Cy = -t \text{ or } y = -\frac{t}{C}$$

To find all slopes w/ a given value, say $y' = 1$, set $C = 1$ 
 $y' = -1$, set $C = -1$ 
etc...

$$y = -t$$
$$y = t$$



everywhere along the red curve the slope is -1

everywhere along the blue curve the slope is +1

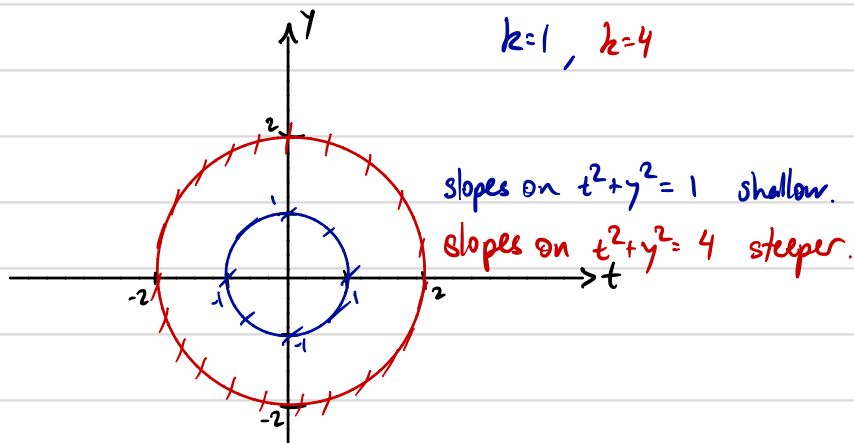
This is a very efficient way to fill in a slope field!

Ex// §1.2 #20, $y' = t^2 + y^2$

To find isoclines, notice $t^2 + y^2 \geq 0$, so no negative slopes exist.

Set $y' = k$, then $t^2 + y^2 = k = (\sqrt{k})^2$

To find slope of value k , we draw circles of radius $\sqrt{k}$.



These sort of qualitative methods reveal things not clear if we only had an analytic solution.

Separability

After getting a qualitative view of the solutions to ODEs, we may try and solve some simpler ones by separating variables.

Defn: A separable ODE is one that can be written $y' = f(t)g(y)$.

Constant solutions can be found by studying $g(y)=0$

Informally, treat $\frac{dy}{dt}$ as a fraction (it isn't, why not?)

Ex// $y' = t^2 y$. This equation is separable since $t^2 y = f(t)g(y)$, w/ const. solutions being $y=0$.

$$\frac{dy}{dt} = t^2 y \Rightarrow \frac{dy}{y} = t^2 dt$$

$$\ln|y| = \frac{1}{3}t^3 + C$$

$$\boxed{y = A e^{\frac{1}{3}t^3}} \quad \leftarrow \text{general solution}$$

Ex// $y' = y^2 + 1$, $\frac{dy}{y^2+1} = dt$, so integrating we find:

$$\arctan(y) = t + C, \text{ or}$$

$$y(t) = \tan(t + C) \quad \checkmark$$

Ex// Solve the IVP $y' = \frac{2t}{1+2y}$, $y(2)=0$

$$\text{Separate: } (1+2y)dy = 2t dt$$

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Separate: $(1+2y)dy = 2t dt$

$$y + y^2 = t^2 + C, \text{ use the IC}$$

$$0 + 0^2 = 2(2) + C, \quad C = -4$$

$\therefore y + y^2 = t^2 - 4$ is the particular solution.

Ex// Some ODEs don't look separable at first, like $\frac{dy}{dt} = f\left(\frac{y}{t}\right)$.

let $u = \frac{y}{t}$, so $y = ut$, so that

$$\frac{dy}{dt} = u + t \frac{du}{dt} \quad (\text{product rule})$$

$$\therefore u + t \frac{du}{dt} = f(u) \quad \text{or} \quad \frac{du}{dt} = \frac{f(u) - u}{t} \quad (!)$$

$$\Rightarrow \frac{du}{f(u) - u} = \frac{dt}{t} \quad \text{and thus separable!}$$

Sub Ex// $\frac{dy}{dt} = \frac{y+t}{t}$ $f\left(\frac{y}{t}\right) = \left(\frac{y}{t}\right) + 1$, so let $u = \frac{y}{t}$, and use the above!

$$f(u) = u + 1$$

$$\frac{du}{f(u) - u} = \frac{dt}{t} \Rightarrow \frac{du}{(u+1) - u} = \frac{dt}{t}$$

Integrating...

$$u = \ln|t| + C$$

$$y = t \ln|t| + Ct$$

↙ back substitute

Picard's Existence & Uniqueness Theorem

In mathematics there is a pair of ideas that occurs time and again, and they usually occur together:

"At Least" and "At Most"

Sometimes we wish to learn if a solution to an ODE exists w/o having solved it. The reason being that we can study properties of those solutions w/o having to deal w/ the actual function.

Sometimes also we need to know if a solution we discover to an ODE is the only one. This might discount unexpected oddities (if multiple solutions existed yet we were aware of only one).

Formally:

Existence theorems give us at least one of some property

Uniqueness theorems give us at most one of some property.

Said in the context of ODEs,

Existence theorems tell us there may be $\{1, 2, 3, \dots\}$ number of sol's

Uniqueness theorems tell us there may be $\{0, 1\}$ number of sol's.

This next theorem is a remarkable example of both an existence & uniqueness theorem

(No proof)

Theorem: (Picard) Given $\frac{dy}{dt} = f(t, y)$, $y(t_0) = y_0$, an IVP

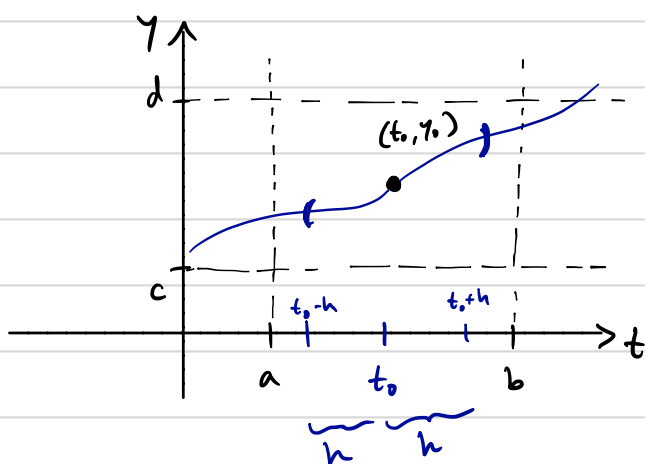
Existence: Suppose the function $f(t, y)$ is continuous on the region

$$R = \{ (t, y) \mid a < t < b, c < y < d \}$$

then $\exists$ a number $h > 0$ s.t. the IVP above has at least one solution for $t \in (t_0 - h, t_0 + h)$.

Uniqueness: If furthermore $\frac{\partial f}{\partial y}$ is also continuous in R , then there is at most one solution.

Remark:



Remark: If f continuous, guaranteed $\{1, 2, 3, \dots\}$ solutions in a neighborhood around the I.C.; and if in addition
If $\frac{\partial f}{\partial y}$ continuous, guaranteed exactly one solution through I.C.!

Ex// $\frac{dy}{dt} = 1 + y^2$. Q: What IC guarantees an existent and unique solution to the ODE?

Note $f(t, y) = 1 + y^2$
 $\frac{\partial f}{\partial y} = 2y$

both are continuous everywhere! Therefore no matter what I.C. we choose, a soln will exist and will be unique!

Ex// #3 §1.5 $y' = y^{4/3}$, $y(0) = 0$

Note $f(t, y) = y^{4/3}$ is continuous around $(0, 0)$, $\therefore$ a solution exists

$\frac{\partial f}{\partial y} = \frac{4}{3} y^{1/3}$ is also continuous around $(0, 0)$, $\therefore$ the solution is unique!

We can solve and find this solution, but that's not the important thing here.

Ex// $y' = \sqrt{y}$, $y(0) = y_0$ (example 3)

$f(t, y) = \sqrt{y}$, this is continuous if $y_0 > 0$ so a solution will exist.

In addition, the curve $y=0$ solves the IVP if $y_0=0$, so a solution also exists in that instance!

However! $\frac{\partial f}{\partial y} = \frac{1}{2\sqrt{y}}$ so as long as $y_0 > 0$ a solution will be unique.

What does Picard's Theorem say about $y(0)=0$?

For $f(t, y) = \sqrt{y}$, there is no open interval around $(0, 0)$ s.t. $\sqrt{y}$ is continuous, so Picard's Theorem does not guarantee the existence of a solution, even though $y=0$ solves the problem!

* This theorem is a sufficiency condition, not a necessity theorem. *

Ex// #26 §1.5 (partly)

Given $\frac{dV}{dt} = k V^{-2/3}$, k positive const., $V(0) = 0$,

Existence: $k V^{-2/3}$ is continuous around $(0,0)$ so Picard's Theorem guarantees a solution.

Uniqueness: $\frac{\partial f}{\partial V} = \frac{2}{3} k V^{-5/3}$ is not continuous at $(0,0)$, so uniqueness is not guaranteed!

We don't know if the IVP has or does not have a unique solution satisfying the IVP! How do we find out?

Manually construct 2 or more solutions. 2 is plenty, really!

① $V = 0$ satisfies the IVP!

② Use separation: $\frac{dV}{V^{-2/3}} = k dt$
 $3 V^{1/3} = kt + C$
 $V = \left(\frac{kt+C}{3}\right)^3$

use IC $(0,0)$
 $0 = \left(\frac{k(0)+C}{3}\right)^3$
 $\Rightarrow C = 0$

$$\therefore V(t) = \left(\frac{kt}{3}\right)^3$$

Since we constructed two solutions that satisfy the IVP, non-uniqueness is proven!